Department of Computer Science

CSE1201: Introduction to Programming with C

Project Specification

Date due: Friday May 3, 2024 at 8pm latest

INSTRUCTIONS

In your main programming assignment for this semester you will create a quiz game application in C in groups of six (6) to eight (8).

The submission date for this coursework is Sunday 7th June 2026 no later than 8pm.

Submission:

By the date/time specified above, you should upload a zip file. The name of the zip file must be in the form:

cse1201-groupnumber-project.zip

e.g. cse1201-group2-project.zip

The zip file must contain your source code, documentation and any resource files needed.

Failure to follow the submission specifications, by providing non-needed files or not following the naming conventions and .zip structure specified, may result in penalised marks.

Plagiarism cases will be handled as documented in the University regulations.

Presentation:

Details about the face to face presentations will be provided in the coming weeks.

Project Concept Specifications

You are to produce a quiz game application in C which is based on a simple question-answer-based project. The quiz game application will enable users to answer a set of questions and get a score for each correct answer. The game should provide users with questions from various topics such as politics, history, geography, current affairs, sports, movies, arts, for example. A Guyanese flavoured game would be a nice touch.

The quiz game application is expected to maintain a leader board and keep track of an individual player’s five highest scores or most recent scores.

You can enhance the game by adding rules, providing bonus questions, requiring a fixed time to answer each question, lifelines, different difficulty levels, etc.

Required features:

- Minimum of 5 questions per game or round
    - Questions can be all from one topic or randomly chosen from the available topics
    - A minimum of 5 questions should be available for about five topic areas.
- Informs user of correct or incorrect answer
- Keeps track of users individual scores (up to 5 scores, most recent or highest)
- Maintains a Leader board of 5-10 highest scores

Optional features:

- Bonus questions with extra points
- Timed answer
    - only providing a specific amount of time to answer 1 question or for a round. For example, 5 secs for each question or 30 seconds for a round or game.
- Lifelines

    - Allowing a specific amount of incorrect answers, each incorrect answer costs one life. So if all lives used, game ends.
    - Alternatively, just keep track of total correct answers.
- Different difficulty levels
- Other

AI Use:

Students are allowed to seek the assistance of a generative AI model e.g. ChatGPT, etc. to assist with coding specific sections of the program. Additionally, they can use AI to assist with outlining the relevant sections for the documentation elements. Any AI use must be reported in detail.

Please note that the use of AI to 'code' the entire game is NOT allowed. Additionally, using AI to fully generate the documentation is also prohibited.

Markscheme:

Criteria	Max Score	Description
Required Functionality (30)		
Basic Functionality	5	Program runs without errors, accepts user's input and presents the options, starts game. Programme ends normally after a round of questions or provides options to continue.
Question feedback	5	Accepts an answer and informs user whether correct or incorrect.
Full game round	10	Presents a minimum of 5 questions to user per game or round. Provides final score to the user
Scoreboard/Leader board	10	Keeps track of an individual's five scores (most recent or highest) and a leader board of top 5-10 scores.
Code Quality (15)		
Good code practises	10	Appropriate use of methods for modularity. Good code technique and good management of memory.
Formatting and commenting	5	Indentation, whitespace, sensible comments throughout the submission
Optional Functionality (10*)		
Two or more optional features	10	Inclusion of two or more optional features
One optional feature	5	Inclusion of at least one optional feature
Presentation (45)		
Project Overview	5	This should describe the game and its game play.
User documentation	10	This should describe how to start and play the game to a user
Technical documentation	20	A document that describes the code and its structure to a technical audience who would possibly want to maintain or modify the code. Present Algorithms, discuss design decisions – logic, data storage mechanisms, code flow. Report AI use.

Presentation Quality	10	This entails how well the presentation is executed, the presentation tool, and how well the group responds in the Q&A. Demonstrated knowledge of submitted programme.

Total Score Calculation:

A group that undertakes the required functionality only can gain a maximum of 90/100 which equates to a score of 90% which would be enough for an A+ for this assessment.

A group that undertakes the required and optional functionality can gain a maximum of 100/100.